EFY TESTED
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Real-Time Environmental DATA LOGGER AND PLOTTER

AKHTAR NADAF

In today's world, where environmental concerns are increasingly at the forefront of global consciousness, the ability to monitor and analyse environmental data has become crucial. From ensuring optimal living conditions in homes and workplaces to tracking climate changes on a larger scale, the importance of real-time environmental data logging cannot be overstated.

The 'real-time environmental data logger and plotter' addresses this need by providing a simple yet effective solution for monitoring temperature and humidity levels in real time. By combining the power of Python programming with the versatility of Arduino microcontrollers, the plotter offers a customisable and accessible platform for environmental data logging. Fig. 1 shows the author's prototype and real-time plotting. The components used to build the device are given in the Bill of Materials table.

Circuit and working

Fig. 2 shows the circuit diagram of the real-time environmental data logger and plotter. It is built around an Arduino Uno or Nano (MOD1) and a DHT sensor (MOD2). Instead of the Arduino Uno, the Arduino Nano can be used with the same pins.

The DHT11 sensor is an affordable and widely available sensor capable of accurately measuring temperature and humidity levels. The Arduino microcontroller interfaces with the DHT11 sensor to capture environmental data, while Python serves as the control centre for data processing, visualisation, and logging.

Data logging is achieved by continuously reading temperature and humidity values from the Arduino via a serial connection. Upon receiving the data, Python processes it and

BILL OF MATERIALS

Components	Quantity
Arduino Nano (MOD1)	1
DHT sensor (MOD2)	1
USB C adaptor	1

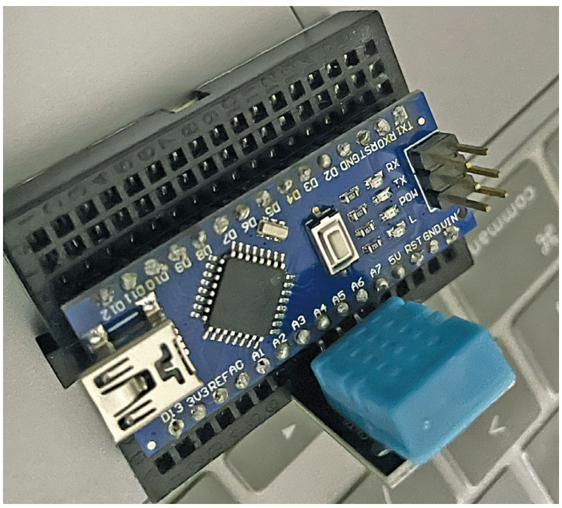

Fig. 1: Author's prototype

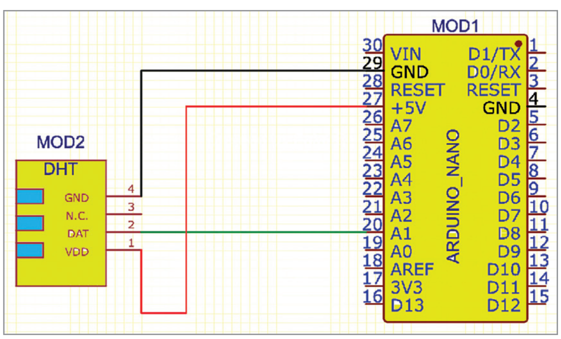

Fig. 2: Circuit diagram

sketch_may7b.ino

```
#include "dht.h"
#define dht_apin A0 // Analog Pin sensor is connected to

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dht DHT;

void setup(){
  Serial.begin(9600);
  delay(500); // Delay to let system boot
  Serial.println("DHT11 Humidity & temperature Sensor\n\n");
  delay(1000); // Wait before accessing Sensor
}

void loop(){
  // Start of Program
  DHT.read11(dht_apin);

  Serial.print(DHT.humidity);
  Serial.print(" ");
  Serial.println(DHT.temperature);

  delay(5000); // Wait 5 seconds before accessing sensor again.
}
```

Fig. 3: Snippet of the source code